Interfacing Python with Excel

In this article, the author discusses interfacing Python with Excel. Data stored in numerous Excel sheets can be extracted by using the Python packages written for the same (like xlrd). And the author's demonstration of its benefits in a real world scenario is enlightening.

Most projects and companies maintain a lot of data in the form of Excel files. These are usually reports, and data needs to be generated from these Excel files in a periodic manner, like a status update. Sometimes, when a specific event takes place, there may be some follow-up action required like generating a mail or filing an escalation in case the data indicates something of interest has happened. So let's look at how such Excel files can be interfaced by using Python.

The problem of work summaries

Let us a look at a specific problem in this regard. Assume that employees in a particular team are working across different projects for different customers. Figure 1 gives an idea about the association between an employee, a project and a customer.

An employee could be working on one or more projects. A project could also have one or more employees. A project is targeted to be delivered to only one customer (in our example) though a customer may be involved with multiple projects.

To track the effort being spent on these projects, employees could be asked to record their work in the following manner under the heads shown below.

Date	Project	Customer	Hours

Let us assume all employees are filing such reports in one Excel file each, in a common location. So how do we arrive at the overall summary of efforts put in by the team on a particular customer, over a period of time?

One of the ways in which this could be managed is to do some fancy linking inside a master Excel file. Then write some Excel macros to walk through each file to summarise the data. As can be imagined, this is tedious, involves manual and automated steps, and can be error prone. Also, summaries over different time periods will have to be managed through macros or filters based on